

Java OOP Concepts Notes: Abstraction, Upcasting & Dynamic Method Dispatch

1. Abstraction

Abstraction means hiding implementation details and showing only functionality.

```
abstract class Animal {  
    abstract void sound();  
}
```

2. Upcasting

Upcasting means referring to a child object using a parent reference.

```
Animal a = new Dog();
```

3. Dynamic Method Dispatch (DMD)

Method call is resolved at runtime based on actual object.

```
Animal a = new Dog();  
a.sound(); // calls Dog's method
```

4. Stack vs Heap

Stack stores references, Heap stores objects.

5. Variables vs Methods

Variables follow reference type (compile time), methods follow object type (runtime).

```
Animal a = new Dog();  
System.out.println(a.type); // Animal
```

6. Practice Programs

```
// Program 1  
Animal a = new Dog();  
a.sound();  
  
// Program 2  
Animal a = new Cat();  
a.sound();  
  
// Program 3  
Animal a = new Dog();  
System.out.println(a.type);
```

7. Key Points

- Abstraction hides implementation - Upcasting enables abstraction usage - DMD resolves method calls at runtime - Variables are not polymorphic - Static methods do not follow DMD